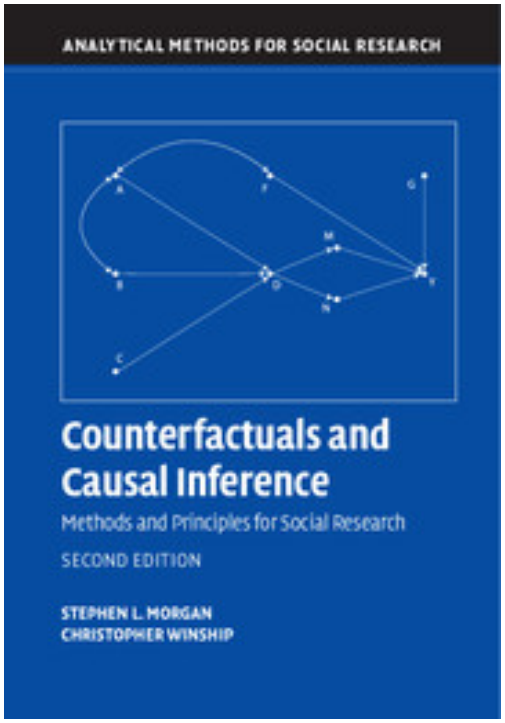
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 8

Outline

1. Nonignorability and Selection on the Unobservables Revisited
2. Selection on the Unobservables and the Utility of Additional Posttreatment Measures of the Outcome
3. Causal Graphs for Complex Patterns of Self-Selection and Heterogeneity

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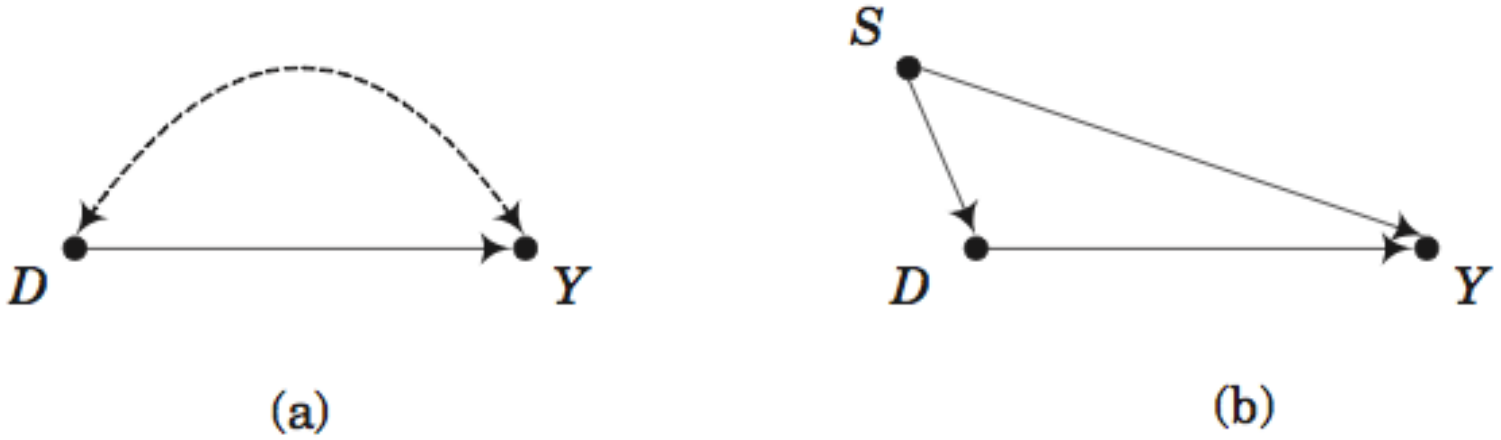
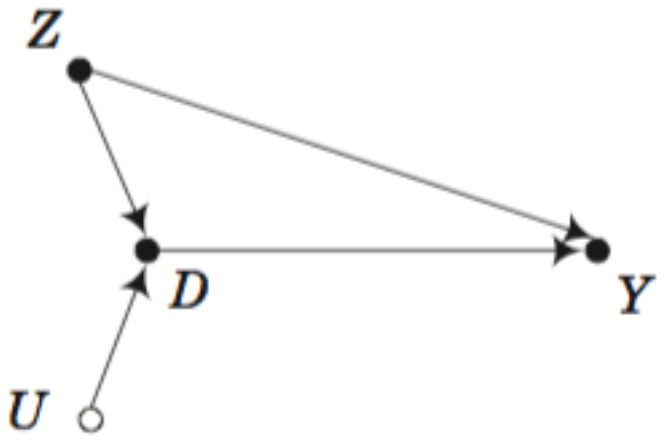
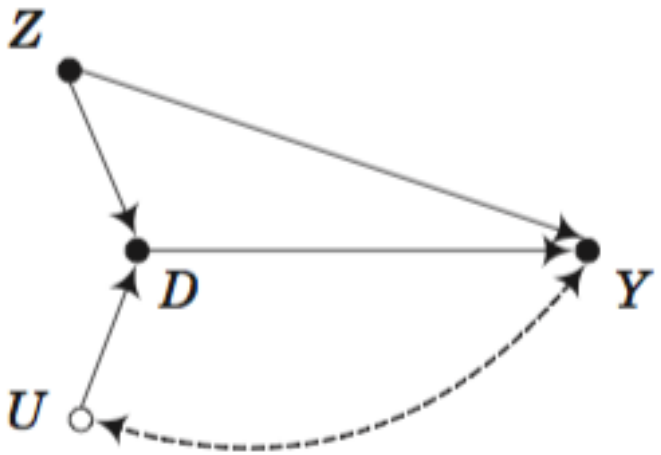


Figure 4.8 Causal diagrams in which treatment assignment is (a) non-ignorable and (b) ignorable.



(a) Selection on the observables



(b) Selection on the unobservables

Figure 4.9 Causal diagrams for the terminology from econometric modeling of treatment selection.

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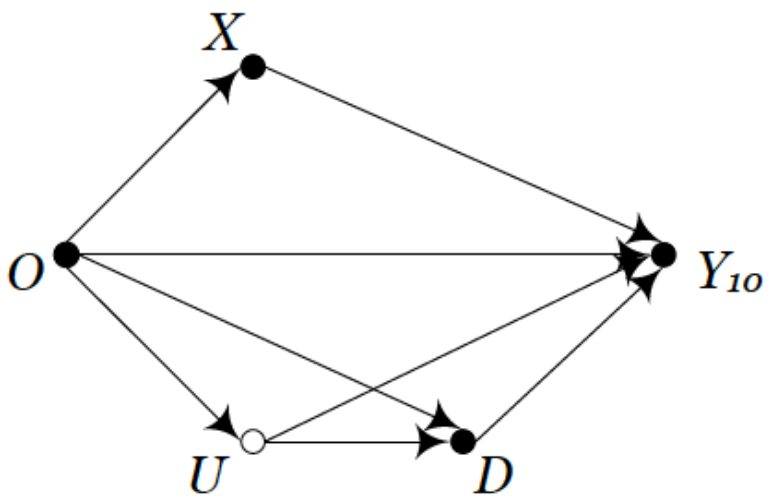
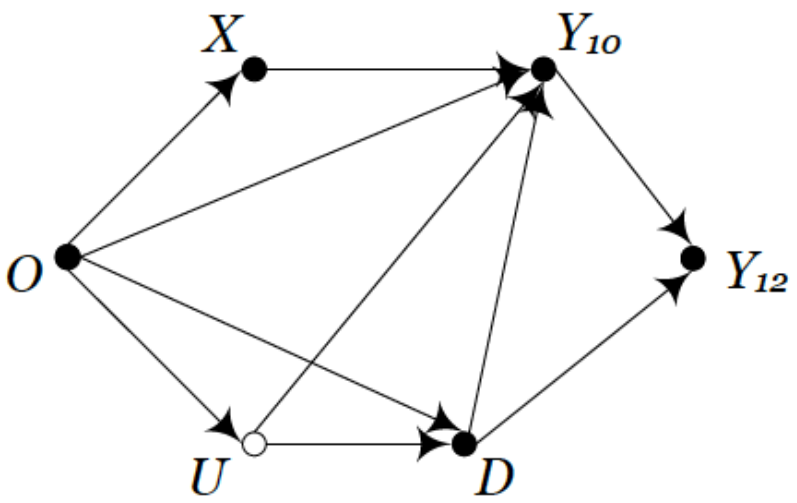


(a) The identification puzzle



(b) Coleman's approach

Figure 8.1 Coleman's strategy for the identification of the causal effect of Catholic schooling on achievement.

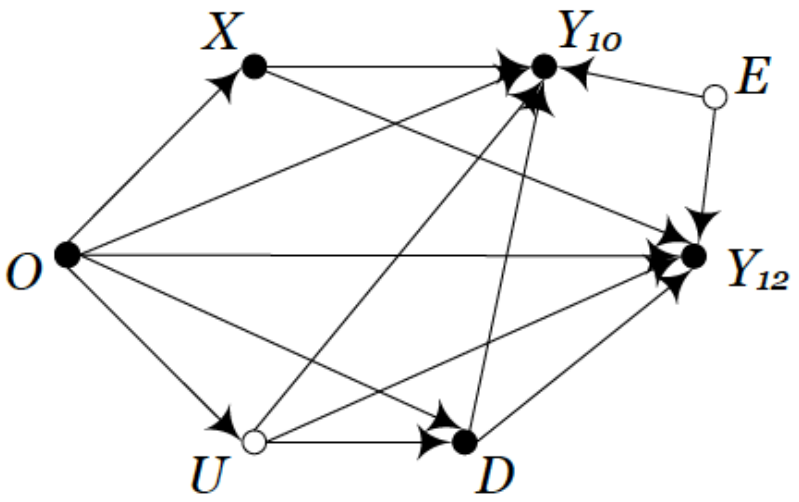


Figure 8.2 Criticism of Coleman's estimates of the effect of Catholic schooling on learning.

Panel Data Demonstration 1

Simplified panel data setup (no difference between D_t and D_t^*):

1. Potential outcome variables are Y_t^1 and Y_t^0, where $t = \{10, 11, 12\}$ for the three grades that occur during the observation window from the tenth grade through the twelfth grade.
2. Because treatment selection occurs before $t = 10$, we have observed data *only* for posttreatment time periods.
3. The treatment indicator variable, D, is equal to 1 if a student is enrolled in a Catholic school and 0 if enrolled in a public school.
4. The observed outcome variable, Y_t, is defined as $DY_t^1 + (1 - D)Y_t^0$ for $t = \{10, 11, 12\}$.

Treatment is determined as in Figures 8.1 and 8.2

- X is a composite determinant of achievement test scores, Y_t, that has no direct effect on whether students select Catholic schooling, D.
- U is a composite variable of unobserved factors that determines both D and Y_t.
- O is a composite variable of ultimate background factors that determines U, X, D, and Y_t.
- The probability of Catholic school enrollment is specified as a logistic distribution

$$\Pr[D = 1|O, U] = \frac{\exp(-3.8 + O + U)}{1 + \exp(-3.8 + O + U)}, \tag{8.1}$$

Two scenarios for Y_t^0 with linear growth

- For the scenarios without supplemental dependence on E,

$$\begin{aligned} Y_{10}^0 &= 100 + O + U + X + \upsilon_{10}^0 \\ Y_{11}^0 &= 101 + O + U + X + \upsilon_{11}^0 \\ Y_{12}^0 &= 102 + O + U + X + \upsilon_{12}^0 \end{aligned} \tag{8.2}$$

  where the υ_t^0 are independent normal random variables with mean of 0.

- For the scenarios with supplemental dependence, Y_t^0 is instead defined as

$$\begin{aligned} Y_{10}^0 &= 100 + O + U + X + E + \upsilon_{10}^0 \\ Y_{11}^0 &= 101 + O + U + X + E + \upsilon_{11}^0 \\ Y_{12}^0 &= 102 + O + U + X + E + \upsilon_{12}^0 \end{aligned} \tag{8.3}$$

Linear growth in the treatment effect, with and without self-selection

- To specify a treatment effect that increases linearly in time, Y_t^1 is defined as

$$\begin{aligned} Y_{10}^1 &= Y_{10}^0 + \delta' + \delta'' \\ Y_{11}^1 &= Y_{11}^0 + (1 + \delta') + \delta'' \\ Y_{12}^1 &= Y_{12}^0 + (2 + \delta') + \delta'' \end{aligned} \tag{8.4}$$

- δ' is a normal random variable with mean of 10 and variance of 1
- In the no-self-selection scenarios, δ'' is a standard normal random variable with mean of 0 and variance of 1
- In the self-selection scenarios, the values of δ'' are set as single draws from individual-specific standard normal random variables with expectation equal to each individual's realized value u_i of U.

Table 8.1 Simulated Results for the Identification Approach Adopted by Coleman and Colleagues

Setup conditions:				
Self-selection on the causal effect	No	No	Yes	Yes
E is a common a cause of Y_t^0	No	Yes	No	Yes
True average treatment effects:				
ATE, tenth grade	10.00	10.00	10.00	10.00
ATT, tenth grade	10.00	10.00	11.85	11.85
ATC, tenth grade	10.00	10.00	9.81	9.81
ATE, twelfth grade	12.00	12.00	12.00	12.00
ATT, twelfth grade	12.00	12.00	13.85	13.85
ATC, twelfth grade	12.00	12.00	11.81	11.81
	Estimated Coefficient on D			
Cross-sectional estimators:				
Regression of Y_{10} on D	14.75	14.75	16.60	16.60
Regression of Y_{10} on O, X, and D	10.80	10.80	12.58	12.58
Regression of Y_{12} on D	16.75	16.75	18.60	18.60
Regression of Y_{12} on O, X, and D	12.80	12.80	14.58	14.58
Panel data analysis of covariance estimators:				
Regression of Y_{12} on Y_{10}, O, X, and D	12.68	9.98	14.41	11.27

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A starting point: separate graphs for separate latent classes

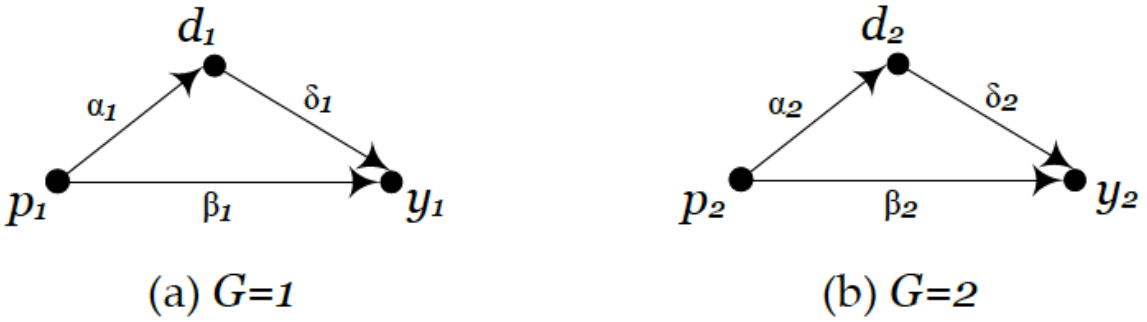

Figure 8.3 Separate causal graphs for two groups of individuals ($G = 1$ and $G = 2$) where the effects of parental background (P) and charter schools (D) on test scores (Y) may differ for the two groups.

A single graph that represents all latent classes

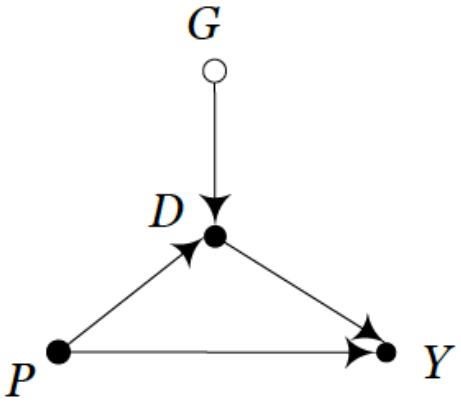

Figure 8.4 A graph where groups are represented by an unobserved latent class variable (G) in a single graph.

Now, the NPSEs ...

- The corresponding structural equations for the graph in Figure 8.4 are

$$P = f_P(e_P) \tag{8.5}$$
$$G = f_G(e_G) \tag{8.6}$$
$$D = f_D(P, G, e_D) \tag{8.7}$$
$$Y = f_Y(P, D, e_Y) \tag{8.8}$$

Selection into latent classes

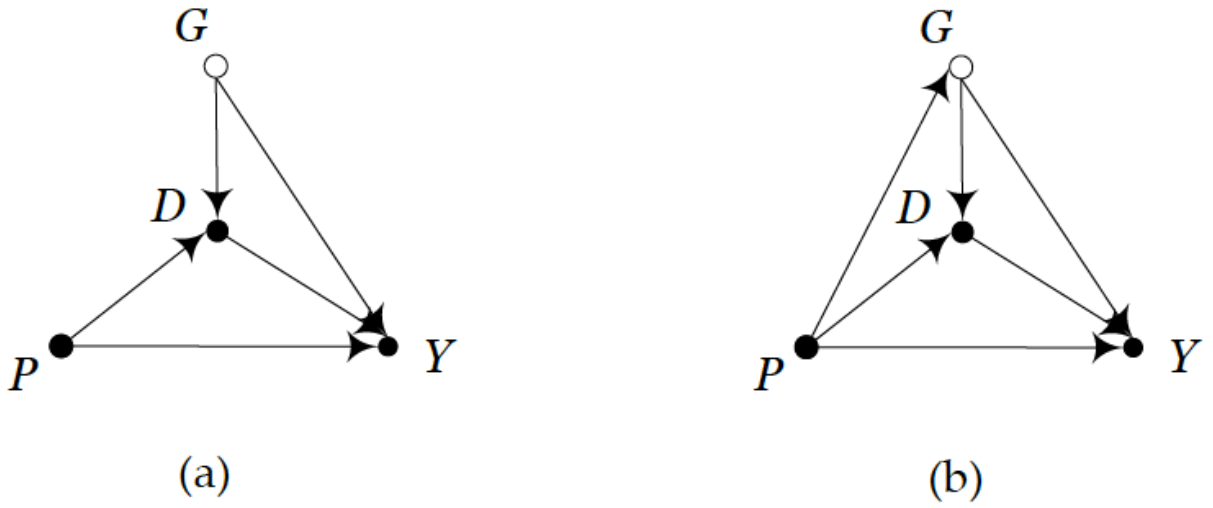

Figure 8.5 Two graphs where selection into charter schools (D) is determined by group (G) and where selection renders the effect of D on Y unidentified as long as G remains unobserved.

- For Figures 8.5(a) we add an arrow from G to Y and the structural equations are

$$P = f_P(e_P) \tag{8.9}$$
$$G = f_G(e_G) \tag{8.10}$$
$$D = f_D(P, G, e_D) \tag{8.11}$$
$$Y = f_Y(P, D, G, e_Y) \tag{8.12}$$

- For Figure 8.5(b), we drop the assumption that G is independent of P. This elaborated graph now includes an arrow from P to G. As a result, $f_G(e_G)$ is no longer the appropriate function for G. Equation 8.10 must be replaced by

$$G = f_G(P, e_G) \tag{8.13}$$

Self-selection based on accurate expectations

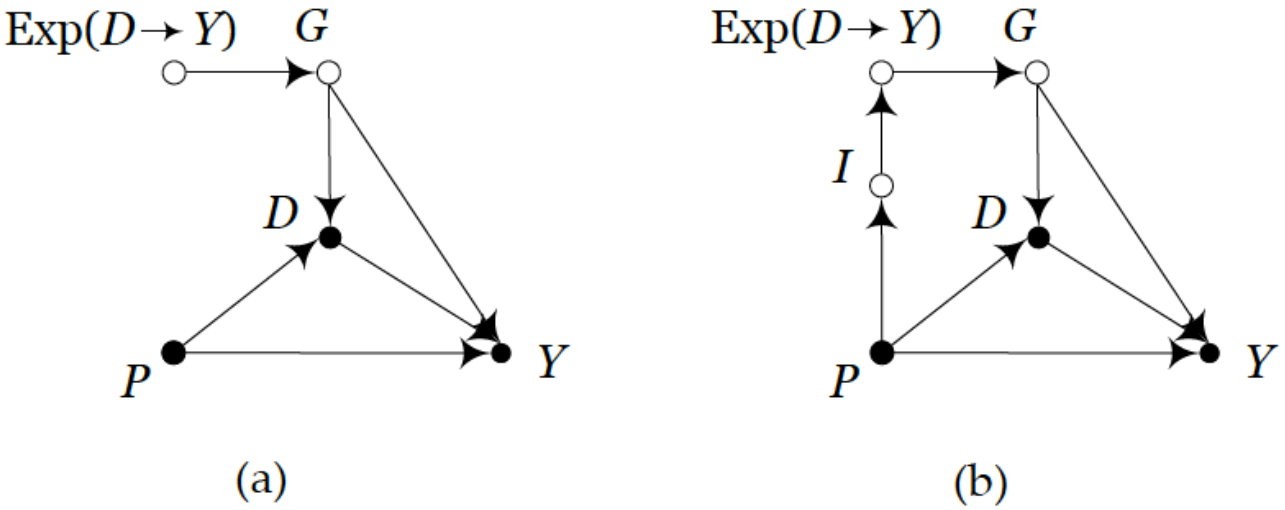

Figure 8.6 Two graphs where selection on the unobservables is given an explicit representation as self-selection on subjective expectations of variation in the causal effect of D on Y. For panel (b), these expectations are determined by information (I) that is differentially available to families with particular parental backgrounds (P).

- For Figure 8.6(a), which is a direct analog to Figure 8.5(a), the subjective expectation is the sole determinant of G. The structural equations are then

$$P = f_P(e_P) \tag{8.14}$$

$$\text{Exp}(D \to Y) = f_{\text{Exp}}(e_{\text{Exp}}) \tag{8.15}$$

$$G = f_G[\text{Exp}(D \to Y),\, e_G] \tag{8.16}$$

$$D = f_D(P, G, e_D) \tag{8.17}$$

$$Y = f_Y(P, D, G, e_Y) \tag{8.18}$$

- For 8.6(b), we have a causal effect of P on $\text{Exp}(D \to Y)$ via I, and t he structural equations are augmented as

$$P = f_P(e_P) \tag{8.19}$$
$$I = f_I(P, e_I) \tag{8.20}$$
$$\text{Exp}(D \to Y) = f_{\text{Exp}}(I, e_{\text{Exp}}) \tag{8.21}$$
$$G = f_G[\text{Exp}(D \to Y),\, e_G] \tag{8.22}$$
$$D = f_D(P, G, e_D) \tag{8.23}$$
$$Y = f_Y(P, D, G, e_Y) \tag{8.24}$$

Self-selection into the treatment and a complementary context

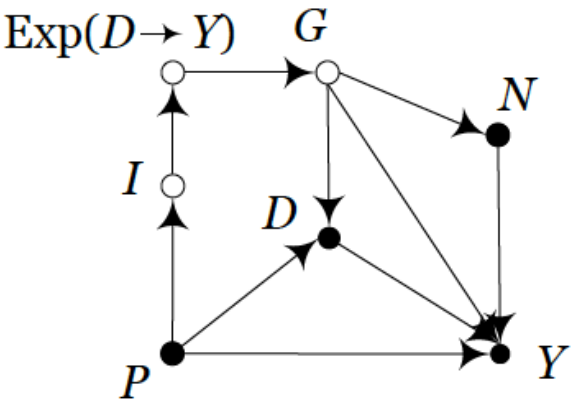

Figure 8.7 A graph where self-selection on the causal effect of charter schooling also triggers self-selection into consequential and interactive neighborhood contexts (N).

- The function for N is equal to $f_N(G, e_N)$, and we then have seven structural equations as

$$P = f_P(e_P) \tag{8.25}$$
$$I = f_I(P, e_I) \tag{8.26}$$
$$\text{Exp}(D \to Y) = f_{\text{Exp}}(I, e_{\text{Exp}}) \tag{8.27}$$
$$G = f_G[\text{Exp}(D \to Y),\, e_G] \tag{8.28}$$
$$D = f_D(P, G, e_D) \tag{8.29}$$
$$N = f_N(G, e_N) \tag{8.30}$$
$$Y = f_Y(P, D, G, N, e_Y) \tag{8.31}$$

A graph-aided interpretation of extant empirical models of charter school effects

- Is there a set of observed variables in Figure 8.7 that satisfies the back-door criterion with the respect the causal effect of D on Y and the causal effect of N on Y?
- The relevant back-door paths are:
  for D,
  1. $D \leftarrow P \rightarrow Y$,
  2. $D \leftarrow P \rightarrow I \rightarrow \text{Exp}(D \rightarrow Y) \rightarrow G \rightarrow Y$,
  3. $D \leftarrow P \rightarrow I \rightarrow \text{Exp}(D \rightarrow Y) \rightarrow G \rightarrow N \rightarrow Y$,
  4. $D \leftarrow G \rightarrow N \rightarrow Y$,
  5. $D \leftarrow G \rightarrow Y$;

- and
  for N,
  6. $N \leftarrow G \rightarrow Y$,
  7. $N \leftarrow G \rightarrow D \rightarrow Y$,
  8. $N \leftarrow G \leftarrow \mathrm{Exp}(D \rightarrow Y) \leftarrow I \leftarrow P \rightarrow Y$,
  9. $N \leftarrow G \leftarrow \mathrm{Exp}(D \rightarrow Y) \leftarrow I \leftarrow P \rightarrow D \rightarrow Y$,
  10. $N \leftarrow G \leftarrow \mathrm{Exp}(D \rightarrow Y) \leftarrow I \leftarrow P \rightarrow D \leftarrow G \rightarrow Y$.
- Combinations of P, N, and D block all back-door paths *except* 5 and 6. The same problematic effect is present for each of them, $G \rightarrow Y$.

What do we need for a complete solution?

- A more elaborate model of G:
  - A model for G, $f_G(.)$, where all inputs other than e_G are observed and where these inputs are sufficiently rich such that e_G can be regarded as a constant that applies to everyone.
  - Equivalently, a set of observed measures as variables that point to G in the graph that would allow us to declare that there are no more latent classes precisely because G is a deterministic function of the observed variables in the graph.
- In this case, all sources of the noncausal association between D and Y and between N and Y would be indexed by the observed variables that determine G, and we could safely remove G from the graph and allow those variables to point directly to D, N, and Y.